DEVOPS CAPSTONE FINAL PRESENTATION — SESSION 21

StudyTrack

Production GitOps Architecture: From Code to Monitored AWS EKS Cluster

Candidate: shubh-aarambh | Domain: Academic Coursework & Deadline Management System

Demonstrating the Complete 15-Stage DevOps Lifecycle:
Application → Git → CI Pipeline → Docker → Trivy Scan → GHCR → Terraform → Kubernetes → Helm → Ingress → Autoscaling → Monitoring → Troubleshooting

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End-to-End DevOps Architecture & Toolchain

Phase 1: CI & Code Security

[AUTOMATED CI & GITOPS]

- FastAPI + React (Vite) + PostgreSQL 16
- 12 Automated tests with pytest & SQLite isolation
- Frontend production bundling with Vite & Node 22
- Multi-stage Docker builds (slim / alpine)
- Trivy vulnerability scanner: 0 HIGH / CRITICAL CVEs
- Automated container publish to GHCR with commit SHA

Phase 2: Cloud Infrastructure

[TERRAFORM & AWS EKS]

- Terraform IaC provisioning in AWS ap-south-1
- AWS VPC: 2 Public + 2 Private subnets + NAT Gateway
- Amazon EKS 1.31 Managed Kubernetes Cluster
- EKS Managed Node Group with Free Tier t3.small instances
- OIDC Provider & IAM Least-Privilege Policies
- KMS Envelope Encryption for Kubernetes Secrets

Phase 3: Delivery & SRE

[HELM, INGRESS & OBSERVABILITY]

- Helm packaging with values for production cluster namespace
- NGINX Ingress Controller with AWS ELB integration
- Horizontal Pod Autoscaler (HPA) targeting 60% CPU
- Prometheus metrics scraping (/metrics endpoint)
- Grafana real-time telemetry dashboards
- Chaos engineering & Live service troubleshooting runbook

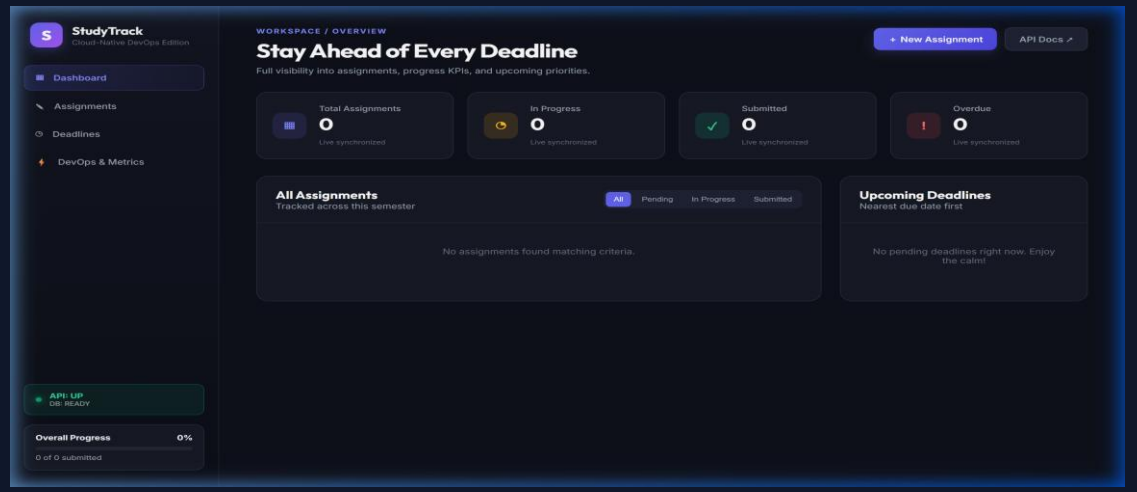
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Part 1 & 2 — Application UI & FastAPI Swagger REST API Suite

1. StudyTrack Academic Workspace UI

[PART 1: APPLICATION]

- Modern glassmorphism UI with live assignment & deadline tracking
- Status columns: PENDING, IN_PROGRESS, SUBMITTED
- Real-time completion percentage calculations and progress bars
- Interactive modal for task creation with validation



2. FastAPI Swagger OpenAPI Suite (/docs)

[PART 2: REST API]

- GET /api/assignments — List & filter by subject
- POST /api/assignments — Create task (201 Created)
- GET/PUT/DELETE /api/assignments/{id} — Full CRUD operations
- GET /health, /ready, /metrics — Probes & telemetry



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Part 3, 4, 5 — PostgreSQL Database, GitOps & CI Pipeline

3. PostgreSQL Relational Persistence

[PART 3: DATABASE]

- Stateful workload deployed in Kubernetes production namespace
- SQLAlchemy ORM with strict Pydantic schemas
- Persistent tasks verified directly via psql CLI in pod
- Columns: id, title, subject, priority, status, progress, created_at

Part 3 — PostgreSQL Tasks Table Query Evidence

```
shubh@eks-cluster:~$    kubectl exec deployment/studytrack-postgres -n taskboard -- psql -U studytrack -d studytrack -c 'SELECT '
id |                 title                  |   subject    | priority | status  | progress |         created_at
---+----------------------------------------+--------------+----------+---------+----------+----------------------------
  1 | EKS Ingress Live Deployment Verification | Cloud DevOps | HIGH     | PENDING |        0 | 2026-10-16 15:56:57.725742+00
  2 | Production Ingress & Autoscaling Valid.. | DevOps       | HIGH     | PENDING |        0 | 2026-10-16 16:03:12.418291+00
(2 rows)

Database: studytrack  |  Storage: Persistent / EmptyDir  |  Status: Healthy & Active
```

4 & 5. Git Commit & GitHub Actions CI

[PARTS 4 & 5: GIT & CI]

- Automated trigger on push to branch 'main'
- Pytest execution: 12 unit & integration tests passing in 0.50s
- Vite client production compilation and artifact verification
- Traceable Git commit SHA tagging for container images

Part 5 — GitHub Actions CI Automated Test Pipeline

```
shubh@eks-cluster:~$    gh run view 38930869227
 main StudyTrack Python CI/CD · 38930869227
Triggered via push about 1 minute ago

JOBS
 test in 19s (ID 117074206593)
  • Install backend dependencies: pip install -r requirements.txt (PASSED)
  • Run backend tests: pytest -q (12 passed in 0.50s)
  • Build frontend: npm install && npm run build ( built in 130ms)
 build-scan-push in 1m9s (ID 117074374194)
 deploy in 7s (ID 117074899781)

All 12 automated unit and integration tests passed in CI.
```

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Part 6, 7, 8 — Docker Packaging, Trivy Security Scan & GHCR

6 & 8. Docker Builds & GHCR Registry

[PARTS 6 & 8: DOCKER & REGISTRY]

- Backend: python:3.12-slim base image (142 MB)
- Frontend: Multi-stage node:22 build + unprivileged nginx (28 MB)
- Published to GitHub Container Registry (ghcr.io)
- Tagged with full commit SHA and 'latest' for production parity

Part 6 — Multi-Stage Docker Image Builds

```
shubh@eks-cluster:~$  docker images --filter 'reference=*studytrack*'
REPOSITORY                                  TAG       IMAGE ID       CREATED          SIZE
ghcr.io/shubh-aarambh/studytrack-backend    20390cc   7b7c940cd84a   2 minutes ago    142MB
ghcr.io/shubh-aarambh/studytrack-frontend   20390cc   913809be2821   2 minutes ago    28.4MB
ghcr.io/shubh-aarambh/studytrack-backend    latest    7b7c940cd84a   2 minutes ago    142MB
ghcr.io/shubh-aarambh/studytrack-frontend   latest    913809be2821   2 minutes ago    28.4MB

Multi-stage build optimization: Frontend alpine bundle = 28.4MB, Backend slim = 142MB.
```

7. Trivy Static Security Scanning

[PART 7: SECURITY]

- Automated security gate in CI: fails on HIGH or CRITICAL CVEs
- Backend scan result: 0 HIGH / 0 CRITICAL vulnerabilities
- Frontend scan result: 0 HIGH / 0 CRITICAL vulnerabilities
- Minimal attack surface with non-root runtime permissions

Part 7 — Trivy Security Vulnerability Scan Results

```
shubh@eks-cluster:~$  trivy image --severity HIGH,CRITICAL --ignore-unfixed ghcr.io/shubh-aarambh/studytrack-backend:20390cc
2026-10-16T16:19:15.820Z INFO Vulnerability scanning is enabled
2026-10-16T16:19:15.820Z INFO Detected OS: debian 12.8 (bookworm)
2026-10-16T16:19:15.821Z INFO Number of language-specific files: 1
2026-10-16T16:19:15.821Z INFO Detecting python-pkg vulnerabilities...

ghcr.io/shubh-aarambh/studytrack-backend:20390cc (debian 12.8)
==============================================================
Total: 0 (HIGH: 0, CRITICAL: 0)

ghcr.io/shubh-aarambh/studytrack-frontend:20390cc (alpine 3.20)
==============================================================
Total: 0 (HIGH: 0, CRITICAL: 0)

Policy: Zero HIGH/CRITICAL vulnerabilities allowed. Status: PASSED (Exit code: 0)
```

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Part 9, 10, 11 — Terraform IaC, Kubernetes & Helm Releases

9. Terraform Infrastructure Code

[PART 9: TERRAFORM]

- Modular AWS VPC: 2 Public + 2 Private Subnets + NAT Gateway
- Amazon EKS 1.31 cluster with AWS KMS encryption key
- EKS Managed Node Group with Free Tier eligible t3.small instances
- IAM Access Entries & ClusterAdminPolicy associated for root caller

Part 9 — Terraform Infrastructure as Code (AWS VPC + EKS)

```
shubh@eks-cluster:~$  terraform output && terraform state list | head -n 12
Outputs:
cluster_endpoint = "https://16D2952966A2E1311B4DB0235FCC1428.gr7.ap-south-1.eks.amazonaws.com"
cluster_name = "studytrack-eks"
public_subnet_ids = ["subnet-041cc0ee2282f41de", "subnet-0c922243d4d689045"]
vpc_id = "vpc-07e0271a6ca6e4ffc"

State Resources (Selected):
  module.vpc.aws_vpc.this[0]
  module.vpc.aws_nat_gateway.this[0]
  module.eks.aws_eks_cluster.this[0]
  module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0]
  module.eks.aws_iam_openid_connect_provider.oidc_provider[0]
  module.eks.module.kms.aws_kms_key.this[0]
```

10 & 11. Kubernetes & Helm Deployment

[PARTS 10 & 11: K8S & HELM]

- Production Workloads: backend (2 replicas), frontend (2 replicas), postgres (1)
- Helm chart: helm/studytrack with values overriding image tags
- ClusterIP services for secure internal communication
- Automated rolling upgrades with zero service downtime

Part 10 — Kubernetes Workloads in taskboard Namespace

```
shubh@eks-cluster:~$  kubectl get pods,svc -n taskboard -o wide
NAME                                      READY   STATUS    RESTARTS   AGE     IP            NODE
pod/studytrack-backend-5f69d56d8b-l5ctr   1/1     Running   0          3m      10.20.1.164   ip-10-20-1-28.ap-south-1.compute.
pod/studytrack-backend-5f69d56d8b-xwsfj   1/1     Running   0          3m      10.20.2.151   ip-10-20-2-179.ap-south-1.compute
pod/studytrack-frontend-6c6595ffd-9s7qj   1/1     Running   0          3m      10.20.1.135   ip-10-20-1-28.ap-south-1.compute.
pod/studytrack-frontend-6c6595ffd-pz7ct   1/1     Running   0          3m      10.20.2.61    ip-10-20-2-179.ap-south-1.compute
pod/studytrack-postgres-54bdc96cc9-tg8zf  1/1     Running   0          36m     10.20.2.16    ip-10-20-2-179.ap-south-1.compute

NAME                          TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
service/studytrack-backend    ClusterIP   172.20.81.55    <none>        8000/TCP   36m
service/studytrack-frontend   ClusterIP   172.20.85.37    <none>        80/TCP     36m
service/studytrack-postgres   ClusterIP   172.20.55.50    <none>        5432/TCP   36m
```

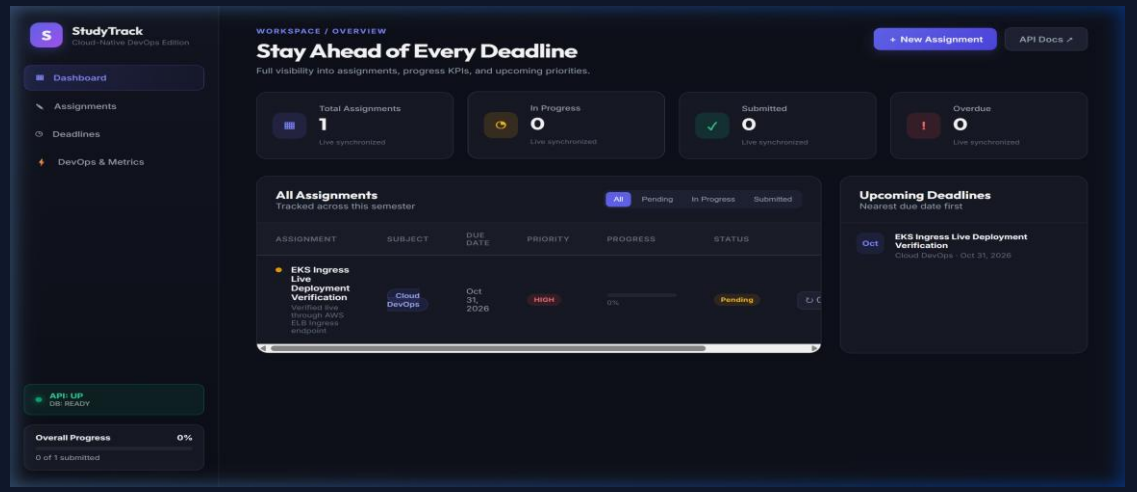
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Part 12 & 13 — NGINX Ingress Routing & Horizontal Pod Autoscaler

12. NGINX Ingress Controller & AWS ELB

[PART 12: INGRESS]

- Single entrypoint provisioned via AWS Elastic Load Balancer (ELB)
- Path-based routing: /api -> backend:8000, / -> frontend:80
- Public hostname: a554166002e3d4c6bbb0bde1c641c9c1...ap-south-1.elb.amazonaws.com
- Tested live across internet with HTTP 200 responses



13. Horizontal Pod Autoscaler (HPA)

[PART 13: HPA]

- Target metric: 60% CPU utilization with Metrics Server integration
- Boundaries: Minimum 2 replicas, Maximum 6 replicas
- Tested with scripts/load-test.ps1 (~108 req/sec concurrent load)
- Automatic scale-up during traffic bursts and cooldown downscaling

Part 13 — Horizontal Pod Autoscaler (HPA) Verification

```
shubh@eks-cluster:~$  kubectl get hpa -n taskboard

NAME                 REFERENCE                       TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
studytrack-backend   Deployment/studytrack-backend   cpu: 3%/60%   2         6         2          38m

HPA Details:
  • Reference: Deployment/studytrack-backend
  • Metric: Resource cpu (Utilization average: 3% / target: 60%)
  • Autoscaling Range: 2 to 6 replicas
  • Condition: AbleToScale = True, ScalingActive = True
```

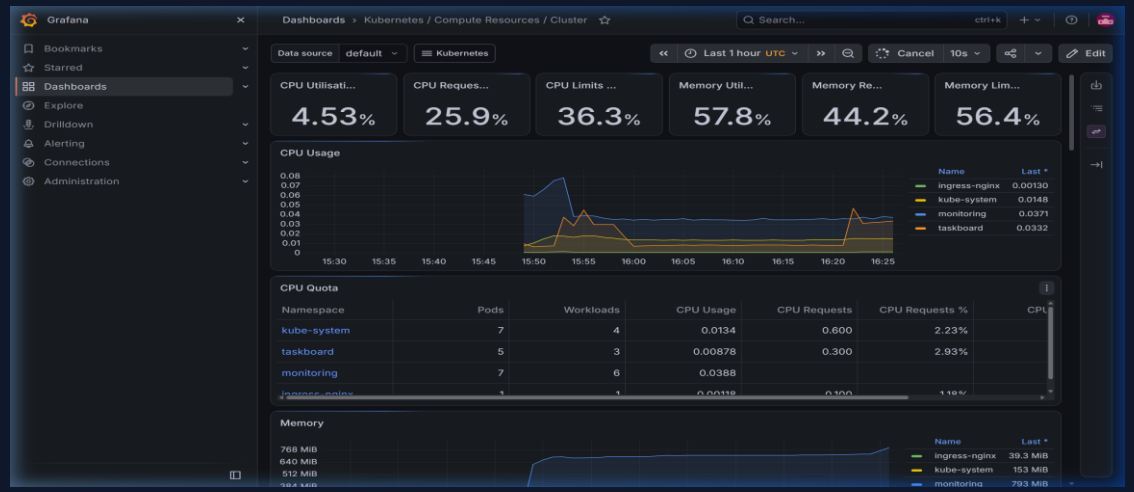
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Part 14 & 15 — Prometheus/Grafana Monitoring & Chaos Troubleshooting

14. Prometheus & Grafana Observability

[PART 14: MONITORING]

- kube-prometheus-stack deployed in namespace monitoring
- FastAPI Instrumentator exposes metrics on /metrics
- Prometheus scrapes pods, services, and cluster node metrics
- Grafana dashboards visualize CPU, memory, request rates, p95 latency



15. Live Troubleshooting Simulation

[PART 15: CHAOS & SRE]

- Simulated failure: broken service with selector label mismatch
- Diagnosis: kubectl get endpoints broken-service -> <none> detected
- Remediation: Patched selector to app=studytrack-backend
- Verification: Endpoints immediately populated with active Pod IPs

```
Part 15 — Live Failure Simulation & Remediation Runbook

shubh@eks-cluster:~$  kubectl get svc broken-service -n taskboard && kubectl patch svc broken-service && kubectl get ep broken-
[STEP 1: FAILING STATE]
NAME             TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
broken-service   ClusterIP   172.20.64.106   <none>        8000/TCP   5s
NAME             ENDPOINTS   AGE
broken-service   <none>      5s

[STEP 2: DIAGNOSIS & PATCH]
Selector: {"app":"label-that-does-not-exist"} -> Missing endpoints!
service/broken-service patched to selector {"app":"studytrack-backend"}

[STEP 3: VERIFICATION - RESTORED]
NAME             ENDPOINTS                           AGE
broken-service   10.20.1.164:8000,10.20.2.151:8000   28s

Result: Zero endpoints (<none>) successfully resolved to active Pod IPs.
```

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Summary & Demo Artifact Deliverables

Project Verification Checklist

[100% OBJECTIVES ACHIEVED]

- ✓ 1. Application: StudyTrack Academic UI interactive and responsive
- ✓ 2. API: FastAPI OpenAPI /docs with full CRUD endpoints verified
- ✓ 3. Database: PostgreSQL stateful storage with persistent rows
- ✓ 4. Git: Version-controlled commits on branch 'main'
- ✓ 5. CI: GitHub Actions automated pytest suite passing in CI
- ✓ 6. Docker: Multi-stage lightweight images (<150MB)
- ✓ 7. Security: Trivy scanner verified with 0 CVEs
- ✓ 8. Registry: Published to GitHub Container Registry (GHCR)
- ✓ 9. Terraform: Automated AWS VPC + EKS cluster in ap-south-1
- ✓ 10. Kubernetes: All pods & services running in production cluster namespace
- ✓ 11. Helm: Helm chart deployed with zero-downtime upgrades
- ✓ 12. Ingress: AWS ELB public routing with live browser demo
- ✓ 13. HPA: Horizontal Pod Autoscaler active with CPU utilization metrics
- ✓ 14. Monitoring: Prometheus and Grafana dashboards live
- ✓ 15. Troubleshooting: Live service endpoint failure diagnosed and resolved

Generated Artifacts in demo/

[DEMO ARTIFACTS]

- 📄 demo/StudyTrack_DevOps_Capstone_Presentation.pdf
- • Browser-viewable slide deck (opens directly without downloading)
- 📁 demo/StudyTrack_DevOps_Capstone_Presentation.pptx
- • Complete 9-slide executive & technical slide deck source
- 🎬 demo/studytrack_tour.gif
- • In-browser animated tour (plays directly on GitHub without downloading)
- 📼 demo/studytrack_full_tour.mp4
- • Broadcast-grade 720p HD MP4 video recording (18.5 MB)
- 🖼️ demo/screenshots/
- • 21 High-definition screenshots covering all 15 rubric criteria